Sallen-Key stage. As described in Chapter 3, the Butterworth filter is maximally flat, that is, it has the sharpest transition between the passband and stopband *without peaking* in the frequency response. The more poles used, the faster the transition. Since all the poles in the Butterworth filter lie on the same circle, all the cascaded stages use the same values for *R* and *C*. The only thing different between the stages is the amplification. Why does this circular pattern of poles provide the optimally flat response? Don't look for an obvious or intuitive answer to this question; it just falls out of the mathematics.

Figure 32-11 shows how the pole positions of the Butterworth filter can be modified to produce the **Chebyshev filter**. As discussed in Chapter 3, the Chebyshev filter achieves a sharper transition than the Butterworth at the expense of ripple being allowed into the passband. In the s-domain, this corresponds to the circle of poles being flattened into an *ellipse*. The more flattened the ellipse, the more ripple in the passband, and the sharper the transition. When formed from a cascade of Sallen-Key stages, this requires different values of resistors and capacitors in each stage.

Figure 32-11 also shows the next level of sophistication in filter design strategy: the **elliptic filter**. The elliptic filter achieves the sharpest possible transition by allowing ripple in both the passband and the stopband. In the s-domain, this corresponds to placing zeros directly on the real axis, with the first one near the cutoff frequency. Elliptic filters come in several varieties and are significantly more difficult to design than Butterworth and Chebyshev configurations. This is because the poles and zeros of the elliptic filter do not lie in a simple geometric pattern, but in a mathematical arrangement involving elliptic functions and integrals (hence the name).

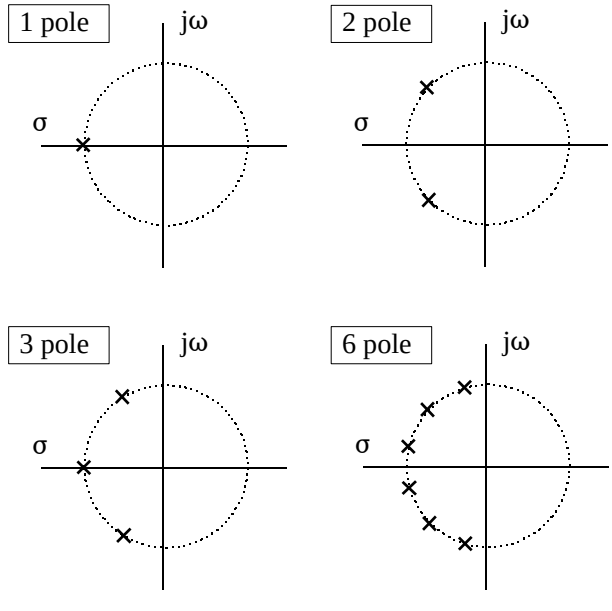

FIGURE 32-10
The Butterworth s-plane. The low-pass Butterworth filter is created by placing poles equally around the left-half of a circle. The more poles used in the filter, the faster the roll-off.